BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

IP and MAC Addresses

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How does wired Ethernet differ from Wi-Fi?

2. What does a server do in a client-server network?
   A) Only requests services
   B) Replaces every client
   C) Provides services to clients
   D) Only connects cables

3. Add 00010111 and 00000110 as unsigned binary numbers. Give the eight-bit result.

4. (a) Logically shift unsigned eight-bit 00110001 one place left. Give the eight-bit result and denary value. (b) Logically shift unsigned eight-bit 00110001 one place right. Give the eight-bit result and denary value.

Today's objectives

Recognise what an IP address and MAC address identify. Explain why switches use MAC addresses and routers use IP addresses.

Key vocabulary - Read, cover, recall

IP address
Network address used to identify and route to a destination.

IPv4
A 32-bit IP addressing format typically written as four decimal octets.

IPv6
A larger IP addressing format written using hexadecimal groups.

MAC address
An address identifying a network interface, typically six hexadecimal pairs.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

IP addresses

An IP address identifies a network destination and allows routers to forward traffic between networks. IPv4 is commonly shown as four decimal numbers.

MAC addresses

A MAC address identifies a network interface; it is commonly shown as six pairs of hexadecimal digits.

Switch and router jobs

Within a LAN, switches forward based on MAC addresses. Between networks, routers use IP addresses.

Compare examples

IPv4 and MAC addresses have different roles and formats. Do not confuse a device's IP address with the MAC address of its NIC.

I Do - Worked model

A computer sends data to a website on another network. Explain why both MAC and IP addressing may be used. [4 marks]

- The computer first communicates on its local network, then needs to reach a remote network.
- Local network interfaces use MAC addresses [1], allowing switching towards the local gateway [1].
- Routers use IP addresses [1] to forward data between networks towards the destination [1].
- Match four points to local MAC use and remote IP routing.

We Do - Guided practice

1. A printer and PC share one LAN. Explain why MAC addresses are useful to a switch. [4 marks]

2. A college connects its LAN to an external website. Explain how IP addresses support the connection. [4 marks]

3. A technician sees 10.2.4.8 and 5A:01:2B:0C:34:EF. Compare these two addresses. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the message printed.

```
octet = 260
if octet > 255 then
print("Outside IPv4 range")
endif
```

You Do - Independent exam questions

Q2. State one role of an IP address. [1 mark]

Q3. A museum ticket scanner sends entry data to the reception computer on the same LAN. Explain how MAC addresses help the switch forward this data. [4 marks]

Q4. Compare IP addressing and MAC addressing and explain their use by routers and switches. [6 marks]

Exit Ticket - True or False?

A router primarily uses destination IP addresses to forward between networks.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Ethernet uses cables. Wi-Fi uses radio, allowing movement but risking interference.

2. C - Provides services to clients

3. 00011101; 23 + 6 = 29.

4. (a) 01100010 = 98. Fill the empty bit with 0. (b) 00011000 = 24. Fill with 0 and lose the old rightmost bit.

I Do - Explain the model

The computer first communicates on its local network, then needs to reach a remote network.

Local network interfaces use MAC addresses [1], allowing switching towards the local gateway [1].

Routers use IP addresses [1] to forward data between networks towards the destination [1].

Match four points to local MAC use and remote IP routing.

We Do - Model answers

1. A network interface has a MAC address [1] identifying it locally [1]. A switch uses MAC addresses [1] to direct traffic to the intended LAN port/device [1].

2. Destination IP address identifies the remote network destination [1]. Routers inspect IP addressing [1] and forward packets between networks [1] towards the correct server [1].

3. 10.2.4.8 is IPv4 [1] shown as four decimal values [1]. 5A:01:2B:0C:34:EF is a MAC address [1], shown as six hexadecimal pairs [1].

Code First - Answer

```
octet = 260
if octet > 255 then
print("Outside IPv4 range")
endif
```

Expected output: Outside IPv4 range

You Do - Self-mark

Q2: Identifies a network destination/used for routing [1].

Q3: The destination computer's NIC has a MAC address [1], which identifies it on the LAN [1]. The switch uses the destination MAC address [1] to forward the data to the appropriate port towards reception [1].

Q4: IP addresses identify network destinations [1] for routing [1], and routers use them to forward between networks [1]. MAC addresses identify local interfaces [1], and switches use them [1] for local forwarding [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.